

Bonus Objective: Revenue Optimization

The band's manager, Rick, wants to maximize profit at V_Gamma (The Snob Pit). He's given

you the following business context:

Venue capacity: approximately 800 seats

Fixed costs per show: ~\$5,000 (venue rental, crew, equipment)

Variable cost per attendee: ~\$8 (security, cleaning, insurance)

Rick has noticed that crowd energy affects whether people stay, buy drinks, and buy merch

Attendance may be affected by both ticket price and crowd energy—but Rick doesn't know exactly how

Rick's request: "Find me the ticket price that makes the most money. I don't care how you

figure it out—just show me the math and justify it. If you tell me 'it depends on the moon'

I'm docking your pay.

Your task:

Derive your own revenue/profit optimization formula — we are NOT giving you the formula

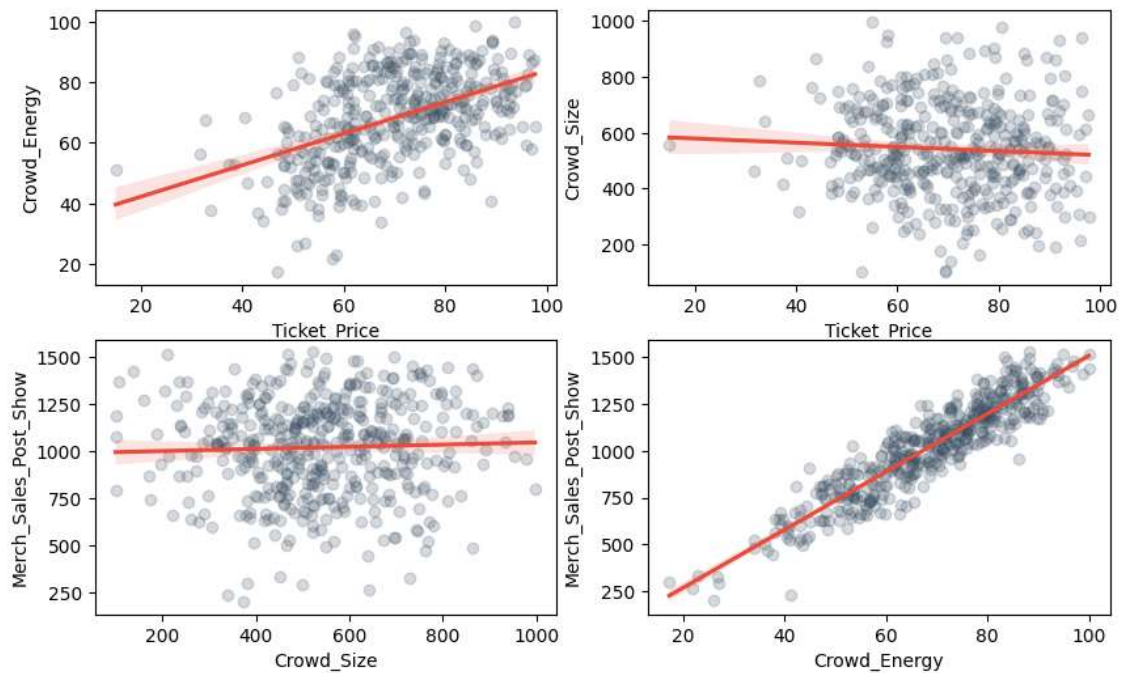
State your assumptions explicitly

Use your trained model to simulate different price points

Create a visualization showing your optimization curve

Identify and justify the optimal price point

My Work:-



I plotted the v_gamma's four plots to find the relationship between them.

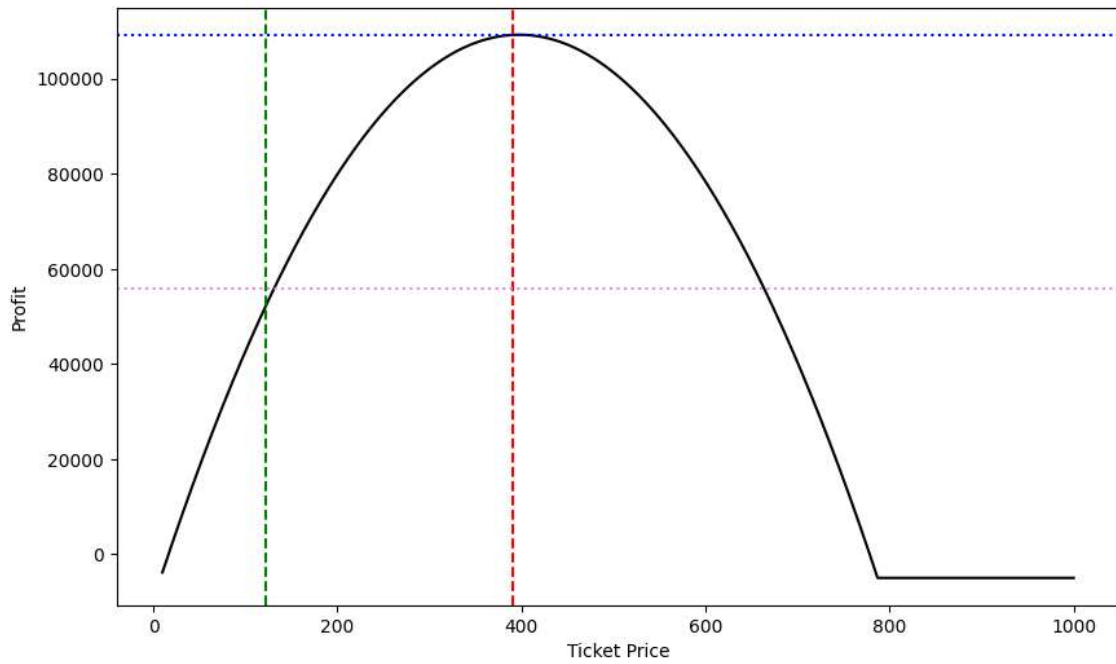
- Clearly merch_sale has pure linear relation w crowd energy.
- I chose merch_sale as function of crowd_energy and crowd_size
- Crowd_size as function of ticket_price
- Crowd_energy as function of crowd_size and ticket_price

Observations:

- Crowd_size isn't much affected by high ticket_price.
- Merch sale post show also isn't affected by crowd_size much.

My profit formula is

$$\text{Crowd_Size} \times (\text{Ticket_Price} - 8) - 5000 + \text{merch_sales_post_show}$$



Applied constraints: Crowd_Size<800, Crowd_Energy <100, merch sale>0

Optimal ticket price is 391(red line) where revenue maximises to 109219.31 usd

For best crowd energy 122(green line) is optimal price with crowd_energy 100